

Chapter 3 — Conditional Statements

AI Study Assistant — RAG Knowledge Base

Learning Objectives

Understand decision-making in Python.

Use if, elif, and else.

Compare values using comparison operators.

Combine conditions with logical operators.

1. Decision Making

Conditional statements allow a program to execute different code depending on whether a condition is true or false.

2. if Statement

The if statement executes its indented block when its condition evaluates to True.

```
age = 20
if age >= 18:
    print("Adult")
```

3. if-else

An else block runs when the if condition is false.

```
marks = 45
if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

4. if-elif-else

elif means 'else if'. It allows multiple conditions to be checked in sequence.

```
score = 72
if score >= 80:
    grade = "A"
elif score >= 60:
    grade = "B"
else:
    grade = "C"
```

5. Comparison Operators

Python provides ==, !=, >, <, >=, and <= for comparisons.

Remember: = assigns a value, while == compares two values.

6. Logical Operators

and requires both conditions to be true.

or requires at least one condition to be true.

not reverses a Boolean result.

```
age >= 18 and has_id == True combines two conditions.
```

7. Nested Conditions

A conditional statement can appear inside another conditional block. This is called nesting.

Nested conditions should be indented carefully so that the intended program structure is clear.

Key Points for Revision

if performs conditional execution.

elif checks another condition when earlier conditions are false.

else handles the remaining case.

== compares; **=** assigns.

and, **or**, and **not** are logical operators.